LAB 20

Capstone: One Command Runs the Pipeline End to End, with a Multi AI Agents Workflow

Item	Detail
Topic	Topic 4: Building an End-to-End Algorithmic Trading Pipeline: Backtest, Forward Test and Performance Analysis
Pipeline stage	End-to-end pipeline
Duration	25 min
Simulator page	http://localhost:8000/#/agents
Mode	No coding. Runs ONLINE with your Alpaca paper keys (Integrations page), or OFFLINE on the demo data and the built-in paper broker. Paper trading only.

Lion City Capital and every scenario in these materials are fictitious and for training only. All trading in this course is paper (simulated) trading. Nothing in these materials is investment advice or a recommendation to trade any security.

Scenario

Lion City Capital's head of trading wants the whole pipeline - data, research, signals, execution, automation, diagnostics and reporting - to run with one command, stop sending orders whenever a gate fails, and finish with a multi AI agents workflow: a lead agent orchestrates finance research, news sentiment, trading strategy, signal generation, a stock trader that places paper orders automatically, performance analysis and a strategy review that feeds the next run.

Objective

Run the whole pipeline end to end, run the 8-agent workflow (research, sentiment, strategy, signal, paper trade, performance analysis, strategy review) and watch its feedback loop, switch on the always-on agent, and turn your own idea into a strategy, then backtest and forward-test it.

Before You Start

- Make sure the Trading Simulator is running and open http://localhost:8000 > Course Labs > Lab 20. It is installed once - see the guide 'Download, Install and Start the Trading Simulator Software' in the Learner Guide.
- Record every result you are asked to check - you will use it in the debrief and in the later labs.

Lab Steps - In the Trading Simulator

No coding is needed. Make sure the Trading Simulator is running (installed once - see the guide 'Download, Install and Start the Trading Simulator Software') and open http://localhost:8000. In the sidebar open Course Labs > Lab 20: the app shows the same steps with a Go button that opens each page pre-filled. Every step works offline on the bundled demo data and the built-in paper broker.

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Step 1. Run the multi-agent workflow

AI Agents > Multi-agent workflow: enter NVDA, AAPL, SPY and press 'Run the multi-agent workflow'. Watch the Agent flow: Lead Agent (plan) -> Finance Research and News Sentiment in parallel -> Trading Strategy -> Signal Generation -> Stock Trader -> Performance Analysis -> Strategy Review.

Open: http://localhost:8000/#/agents (AI Agents > Multi-agent workflow)

Check: The run finishes with 8 agents per symbol (lead, performance analysis and strategy review once per run); the brain badge shows which provider, the rule brain or deterministic code answered.

Step 2. Read each agent's card

Pick each symbol tab and read the eight cards in flow order: the lead's plan and summary; Finance Research (1-min / 15-min / 1-hour / daily candles, Yahoo Finance ratios, Equibles data); News Sentiment (Reddit and NewsAPI scores and notable items); Trading Strategy (the algorithm votes computed in code - trend, momentum, volatility, VWAP, RSI, MACD - and the chosen weights); Signal Generation (long, short or null); Stock Trader; Performance Analysis; Strategy Review.

Open: http://localhost:8000/#/agents

Check: For one symbol you can explain how the composite score became the signal and which algorithms drove it.

Step 3. Check the order on Alpaca paper

Read the Stock Trader card and the 'Paper orders sent' table: long = buy, short = sell short (an opposite position is closed first), null = no order. Orders are placed automatically - there is no approval step. Open Paper Trading > Orders (source agent) and, with Alpaca keys, your Alpaca paper dashboard.

Open: http://localhost:8000/#/trading?tab=orders

Check: Every long / short signal has a matching paper order (or a recorded broker error, e.g. a symbol that cannot be shorted); null signals have none.

Step 4. Read the performance panel and the strategy review guidance

AI Agents > Performance: total return, daily returns, Sharpe, max drawdown, win rate, trades and exposure of the agents' own paper trading (also on the Dashboard). Then AI Agents > Strategy guidance: the weights, thresholds and notes the Strategy Review Agent wrote.

Open: http://localhost:8000/#/agents?tab=performance

Check: You can explain each metric and say which algorithms the review now prefers and why.

Step 5. Run it again to see the feedback loop

Run the workflow again on the same symbols. On the Trading Strategy card check 'Guidance from the Strategy Review of run #...' and compare the weights and thresholds with the first run; Strategy guidance > history shows each review.

Open: http://localhost:8000/#/agents

Check: The second run's Trading Strategy agent used the guidance written by the first run's review, and a new review was saved.

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Step 6. Always-on agent

AI Agents > Always-on agent: enable it with a 60-minute wake-up (the worker must run), or press 'Wake it now'. Check Memory.

Open: http://localhost:8000/#/agents

Check: A new always_on run and memory notes (signals, decisions, guidance) appear.

Step 7. Turn your own idea into a strategy

Idea > Strategy: describe your own trading idea in one or two sentences (signal, when to buy, when to sell) and press 'Turn my idea into a strategy'. Read how the app understood it, then press 'Test it now'. No coding - the app writes and checks the code.

Open: http://localhost:8000/#/builder (Idea > Strategy)

Check: A READY strategy with a PASS look-ahead check, your rules listed in plain English, and a backtest result.

Step 8. Backtest, diagnose and forward-test it

Backtest it, run Diagnostics, then create a paper bot for it and start it.

Open: http://localhost:8000/#/trading?new=1

Check: Pipeline Map > Strategy lifecycle for your strategy shows the gates it passed.

Step 9. End-to-end on the Pipeline Map

Open the Pipeline Map and press 'Run a bot cycle'. Then switch to 'Strategy lifecycle'.

Open: http://localhost:8000/#/pipeline (Pipeline Map)

Check: All 14 stages have run; you can walk a colleague through each stage.

Expected in the app: A Pipeline Map with every stage green, an agent run with its 8-agent trace, a paper order on Alpaca (or the simulator), the agents' performance metrics and strategy guidance, and your custom strategy trading on paper.

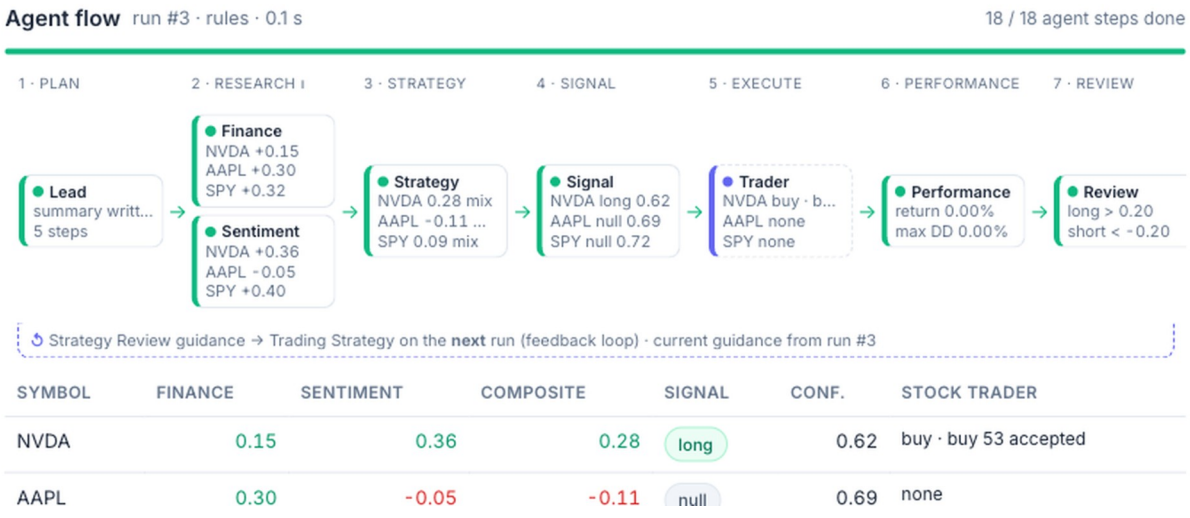


Figure: Lab 20 in the Trading Simulator.

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Optional: Let AI Write the Code

This lab needs no coding. To see, run or customise the code behind it, click a link: the AI opens with the prompt filled in and writes the code for you. All prompts are also in labs/Prompt_Library.html and .pdf.

Your own strategy for the Strategy Builder - paste it into the Strategy Builder
Write a strategy for my Trading Simulator's Strategy Builder. Rules: one Python class that subclasses Strategy (from trader.strategies.base import Strategy, enter_exit, flat_until_ready); import indicators with `from trader.core import indicators as ta` (ta.sma(series, window), ta.ema(series, span), ta.rsi(close, window), ta.macd(close, fast, slow, signal) returns columns macd/signal/hist, ta.bollinger(close, window, k) returns columns lower/middle/upper/pct_b, ta.atr(df, window), ta.donchian(df, window), ta.realized_vol(close, window)); give it a name and a params dict such as {"window": {"default": 20, "min": 2, "max": 200, "help": "Lookback in bars"}}; implement generate_signals(self, data) where data is an OHLCV DataFrame, and return a Series of 1 (long), 0 (flat) or -1 (short) aligned to data.index. Use only data up to each bar - no center=True and no shift(-1) - because the engine trades on the next bar. Reply with the code only, then 3 bullet points explaining the rule. My idea: buy SPY when RSI(14) falls below 30 while the close is above its 200-day SMA, and sell when RSI(14) rises above 55. (Replace this idea with your own.)

Open in Claude | **Open in ChatGPT**

Debrief

Agentic AI chains specialised agents that research, decide and act, which makes it powerful and risky. In this capstone the eight agents trade the paper account automatically - there is no approval step and no agent risk limit - so the controls that remain matter: paper-only execution, the kill switch, market-session rules, duplicate-order protection and a strategy review that learns from measured performance. Before any real money you would add position and loss limits and human sign-off, and evaluate every stage's evidence critically.

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